

113A Introduction to Computer and Computer Science

Homework Assignment #7

Due: 11/11 12:00:00

In this homework assignment, you will practice how to apply the concept of object-oriented programming to solve a practical issue. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

Problem #1: Create a Dice (30%)

There are several iconic games in casino, at least one of them is related to a simple, tiny, but elegant element - dice. In this homework, we are going to create a “**Dice**” object to represent a fair dice, which has several instance and method attributes:

1. An instance attribute “**current_state**” to represent its current point. The initial state should be “None”.
2. An instance attribute “**dice_id**” to represent its identity (btw, how do we check the identification of each dice during the game?).
3. There are many kinds of dice in the world (e.g., cubic dice, D8, D12, ...). You should remain a parameter for the users to change the dice.
4. A method attribute “**roll**” to randomly change the “**current_state**”. For example, the state should be 1, 2, ..., or 6 for a cubic dice with uniform probability density function.
5. A method attribute “**get_current_state**” to access the instance attribute “**current_state**”.

If everything goes well, you might have this result.

```
dice1 = Dice(1, 12)
dice2 = Dice("A", 20)

print(dice1, dice2)

for i in range(5):
    dice1.roll()
    dice2.roll()
    print(dice1, dice2)
```

✓ 0.0s

```
Dice 1: None Dice A: None
Dice 1: 10 Dice A: 15
Dice 1: 1 Dice A: 1
Dice 1: 8 Dice A: 11
Dice 1: 1 Dice A: 19
Dice 1: 2 Dice A: 17
```

Here is the sample code:

```
import random

class Dice:
    """A fare dice"""
    def __init__(self, id, dice_type=6):
        self.current_state = None
        self.dice_type = ???
        self.dice_id = ???

    def __str__(self):
        return ???

    def roll(self):
        self.current_state = ???

    def get_current_state(self):
        return self.current_state
```

!!!NOTICE!!!

In this homework, you are only allowed to
import the module to implement the uniform probability.
(the random module as shown in the above sample code)

Problem #2: A Person with a Dice box (70%)

Let us assume that we have a special guest, Chung Bo-Eat, in our casino, holding a dice box with M dices. Please create a “**Person**” object to represent this person, which also has several instance and method attributes:

1. An instance attribute “**name**” to represent her/his identity.
2. An instance attribute “**dicebox**” to represent the dice box holding on her/his hand.
3. A method attribute “**put**” to put M dices into her/his dice box. She/he should know what kind of dice she/he is going to play.
4. A method attribute “**check**” to check the status of her/his dice box.
5. A method attribute “**roll**” to represent the event of she/he rolling the dice box. This method should change the state of every dices in her/his dice box.
6. A method attribute “**check_sum**” to calculate and return you the summation of all dices in her/his dice box

Here is the sample code:

```
class Person:
    """A person holding a dice box with arbitrary dices"""
    def __init__(self, name, numDices, dice_type=6):
        self.name = ???
        self.dicebox = ???
        self.put(numDices, dice_type)

    def put(self, numDices, dice_type):
        ???

    def roll(self):
        ???

    def check(self):
        ???

    def check_sum(self):
        ???
```

Please accomplish this homework with an organized code (e.g., with main script and function script). Your main script file name must be 'main_hw7.py.' Here is a template for your code structure:

```
113A_hw7_0123456789
├─ obj.py           # Module 1 for hw7
├─ func.py          # Module 2 for hw7
├─ :                # Other modules if necessary
└─ main_hw7.py      # Main script of hw7
```

FRIENDLY REMINDER

You don't need to follow this structure exactly, just keep your main script clean.

Here is a template for your main script.

```
if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="2024 NYCUDOPCS
Homework #7")
    parser.add_argument("--num", default=6, type=int, help="Number of
dices (default: 6)")
    parser.add_argument("--dtype", default=12, type=int, help="Dice type
(default: 12)")

    opt = parser.parse_args()

    test_hw7(numDices=opt.num, diceType=opt.dtype)
```

If everything goes well, you might have this result.

```
(nycudopcs) 0:\                               113A\HW7\113A_hw7_0123456789>python main_hw7.py --num 7 --dtype 87
Game 1:
  Player Hsiao Bo-Bo:
    State of dicebox: [23, 24, 73, 36, 51, 84, 25]
  Player Wei Tung-Tung:
    State of dicebox: [87, 66, 63, 51, 6, 60, 36]
=====
Game 2:
  Player Hsiao Bo-Bo:
    State of dicebox: [9, 55, 18, 10, 83, 45, 69]
  Player Wei Tung-Tung:
    State of dicebox: [65, 43, 3, 52, 35, 14, 30]
=====
```

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
""  
Created on Wed Aug 7 01:23:45 2024  
  
@author: Xi Winnie, student ID  
  
@collaborators: William Lai, his student ID  
""
```

Please save your homework as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

113A_hw7_**ID**.zip

For example,

113A_hw7_0123456789.zip

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature in main script.** Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the “E3@NYCU” system. There will be a section for uploading your homework.